GEOX SABAH REGIONAL

GEOLOGICAL INTELLIGENCE

East Sabah → Dangerous Grounds · Regional Earth system / Basin domain / Local calibration

PETRONAS INTERNAL · LOCAL ONLY · INTERPRETATION · PARTIAL · NOT SEALED

AI_IMAGE_GENERATION: NOT AVAILABLE · EXTERNAL_IMAGE_GENERATION: BLOCKED_C28 · AI-READY CONCEPTUAL VISUAL LAYER · CURRENT IMPLEMENTATION: GOVERNED PROCEDURAL RENDER

Build correctness is not Earth truth. GEOX renders; ARIF adjudicates.
No model is canonical, preferred or closed. P, L and R are drawn with equal weight.

THREE NESTED SCALES

SCALE 1 · Regional Sabah — East Sabah → Dangerous Grounds (this transect)
SCALE 2 · Basin / structural domain — Sabah Trough, toe thrust, thrust sheet, inboard, platform (BIF-KL6)
SCALE 3 · Local calibration — KB-KL2 Rotan–Maligan–Barton

Sources: SRC-XS-PUB · SRC-BIF23-KL6 · v4.2 chronostrat · v4.1 RHG payload · v3.1 surface system (SRC-REST15) · SRC-ESDG-MODELS (hypotheses)

EXECUTIVE INTERPRETATION RAIL

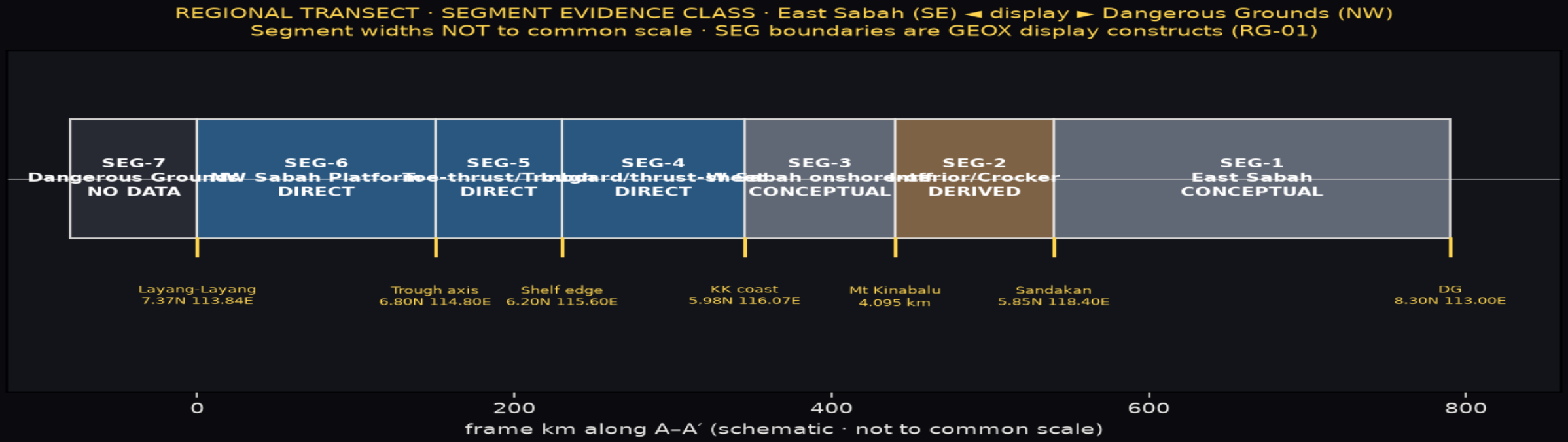
No single grand narrative: competing models are listed side by side; nothing here is adjudicated.

DIRECTLY SUPPORTED	One composite seismic display (BIF-KL6) over NW Sabah Platform, Sabah Trough, toe-thrust and thrust-sheet domains: rugose platform seabed, flat trough floor ≈3800 ms, seabed rising SE toward shelf, offset reflectors in toe-thrust panel. Navigation, datum and depth absent.
LOCAL TO KB-KL2	Digitised horizons · Rotan-1 / Maligan-1 / Barton-2 ties (C20: Rotan SRU/H3 FAIL, Maligan SRU FAIL) · Block H operator horizons (H-series, H400 age unassigned) · restoration-pack chronology behind EVENT-C-S.
MAY BE REGIONAL	Shallowing-to-SE seabed trend and trough → toe-thrust → thrust-sheet → shelf domain order, visible on BIF-KL6, consistent with public A–A′ frame order.
REMAINS A CORRELATION	DRU, SRU and H-series continuity from KB-KL2 onto BIF-KL6 · MMU vs DRU across the trough (C9) · KL6 panel positions relative to frame A–A′.
REMAINS A HYPOTHESIS	EVENT-C-M (Sabah Trough tectono-magmatic branch) · crustal models R1–R4 · Models A/B/C of SRC-ESDG-MODELS · onshore–offshore magmatic correlation · 'Mesozoic' attribution of deep SE section.
NOT REPRESENTED BY DATA	East Sabah and interior Sabah subsurface · mapped surface geology along transect · Dangerous Grounds below seabed · crust and Moho everywhere · depth everywhere.

WHAT CHANGES EXPLORATION UNDERSTANDING: KB-KL2 cannot be used as the regional reference — its failed ties stay local. Regional play statements need a navigated dip keyline across the trough tied to wells. (Consequence statements = interpretation, not fact.)

REGIONAL TRANSECT · SEGMENT EVIDENCE CLASS

EAST SABAH (SE) ◄ DISPLAY ORIENTATION ► DANGEROUS GROUNDS (NW)



Station order in SRC-XS-PUB: Dangerous Grounds (8.30 N, 113.00 E) → Sandakan Basin (5.85 N, 118.40 E).
Display reverses frame km so East Sabah is on the LEFT. BIF-KL6 is NW→SE as sourced; panels shown unflipped, each with a reversal marker.

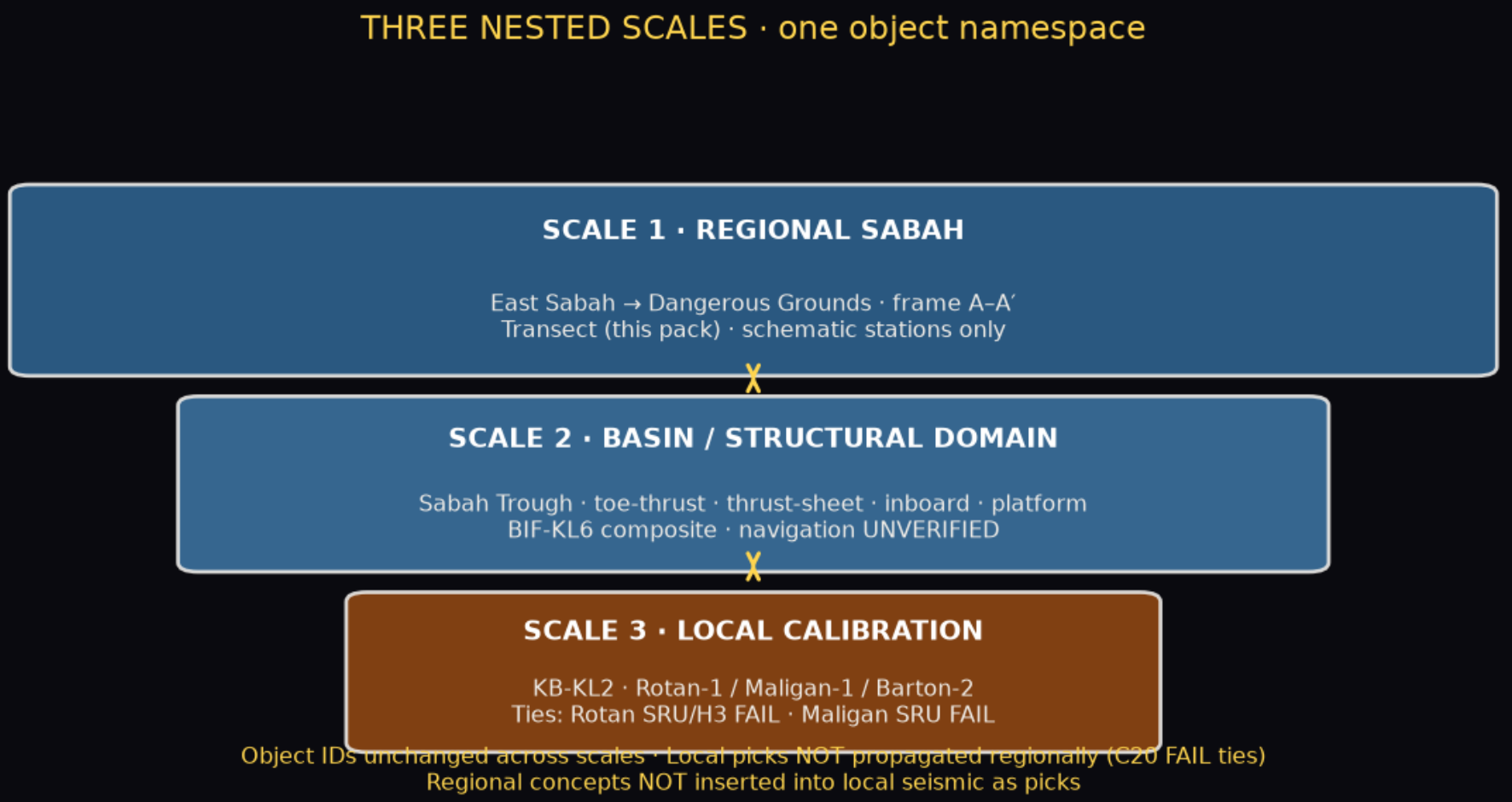
Segment widths on the display are NOT to a common horizontal scale. Segment km ranges are GEOX display boundaries constructed on the public schematic A–A′ frame (RG-01); BIF-KL6 panels are placed by structural-domain name, not by coordinates.

SEGMENT-BY-SEGMENT · WHAT EACH ONE CONTROLS

SEG-1 · East Sabah **CONCEPTUAL BRIDGE**	Geological-map control. SRC-XS-PUB stations (Telupid, Kinabatangan, Sandakan Basin). SB505 / Sandakan / SB304 documents inventoried, NOT ingested. Unknowns: mapped surface geology along transect; subsurface geometry; seismic. Confidence: LOW. Gate: RG-01.
SEG-2 · Interior Sabah / Crocker–Kinabalu **DERIVED**	SRC-XS-PUB topographic nodes (Mt Kinabalu 4.095 km as quoted). SRC-BB-V5 cross-check (secondary). RC-06: granite root in secondary schematics NOT adopted. Unknowns: surface geology; crustal geometry; emplacement depth/shape of Kinabalu body. Gate: RG-02.
SEG-3 · Western Sabah onshore-to-offshore **CONCEPTUAL BRIDGE**	SRC-XS-PUB (Kota Kinabalu coast station). No seismic, no well. Unknowns: any seismic or well tie across the coastline. Confidence: LOW. Gate: RG-01.
SEG-4 · Inboard offshore / thrust-sheet **DIRECT**	SRC-BIF23-KL6 px 1130–1590 ('Thrust Sheet'). TWT ms 0–10000. Wells: Dudar, Tekuyong, Kandi, Sugut, Solisip. Navigation UNVERIFIED. Gates: C27, RG-03.
SEG-5 · Toe-thrust / Sabah Trough **DIRECT**	SRC-BIF23-KL6 px 393–1130 ('NW Sabah Trough', 'Toe Thrust') + KB-KL2 local calibration (v4.1). Wells: Permai, Buluh, Rotan. C9/C20/C27/C30 open. EVT-C-M corridor has NO geometry.
SEG-6 · NW Sabah Platform **DIRECT**	SRC-BIF23-KL6 px 63–393. Rugose seabed ≈1900–2200 ms. RC-03: 'NW Sabah Platform' not in SRC-XS-PUB province list — boundary is a GEOX display construct. Gates: RG-03, RG-04.
SEG-7 · Dangerous Grounds **NO DATA**	SRC-XS-PUB stations only (Dangerous Grounds, Layang-Layang; schematic bathymetry nodes). Nothing below seabed. Confidence: UNKNOWN. Gates: C9, RG-04.

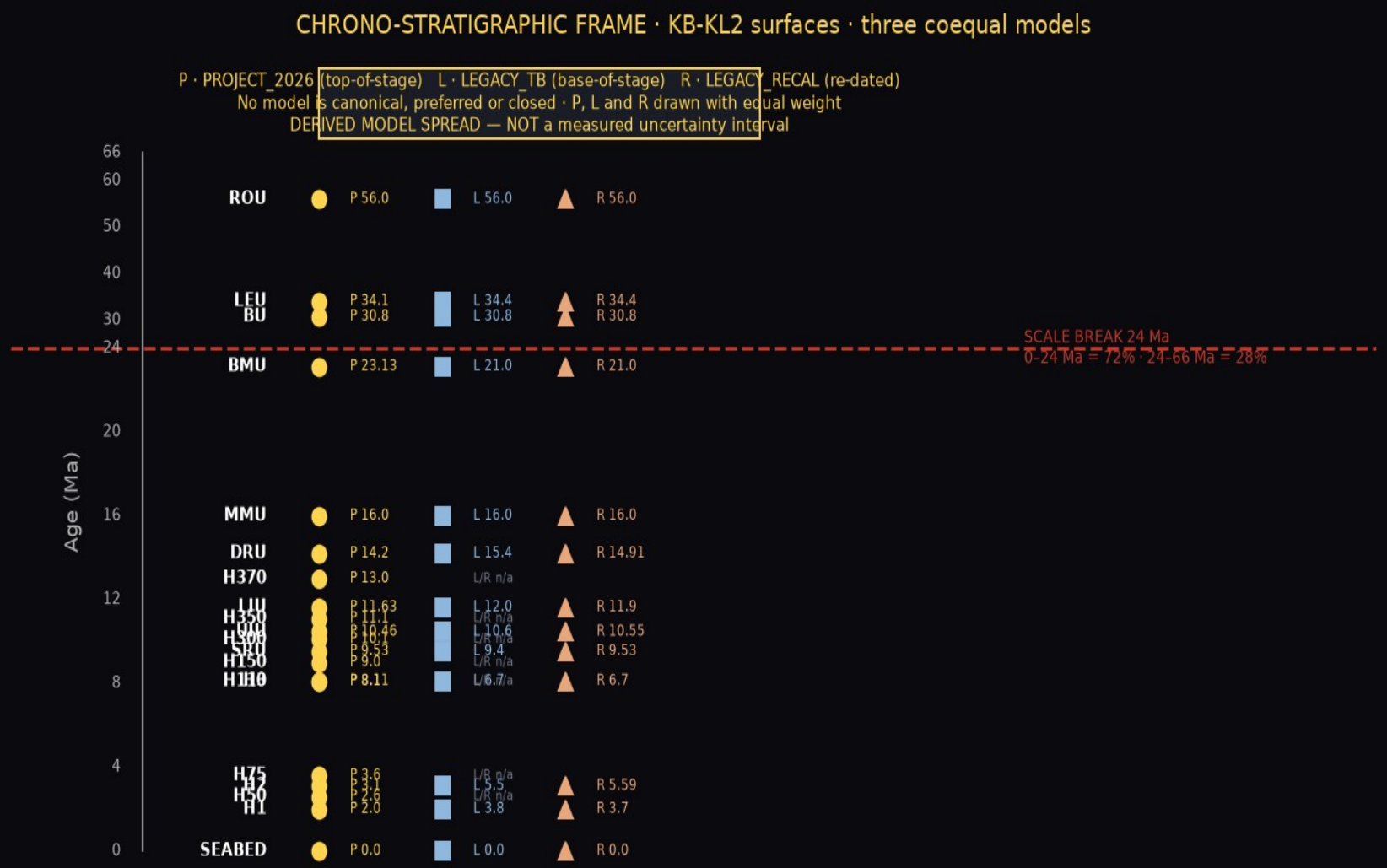
THREE NESTED SCALES

Object IDs unchanged across scales · local picks NOT propagated regionally (C20 FAIL ties)



CHRONO-STRATIGRAPHIC FRAME · KB-KL2 SURFACES

SCALE BREAK 24 Ma · 0–24 Ma = 72% of height · 24–66 Ma = 28% · piecewise linear · P, L and R drawn with equal weight



MODEL STATUS

P · PROJECT_2026
top-of-stage · SRC-UNI, SRC-P1 not local

L · LEGACY_TB
base-of-stage · SRC-PEN local

R · LEGACY_RECAL
re-dated legacy · hypothesis by construction

DERIVED MODEL SPREAD
NOT a measured uncertainty interval

Age not assigned: H400, H136, H120, U-*

H400 → review card · NO AGE · UNSET

GTS2020 stage column + NN (GTS2012)
TS (interpretation) TTSS00–TTSS55

SURFACE CALLOUTS · KB-KL2

P / L / R ages where assigned · restoration-pack quotations marked · confidence per surface

SEABED	P 0.0 · L 0.0 · R 0.0	INTERP LOW-MED · P-gap 0g · C20 UNSET · C20
H50	2.6 Ma · REST15	INTERP LOW-MED · REST15 0g · UNSET
H1	P 2.0 · L 3.8 · R 3.7	INTERP MED · P-gap 1g · C6
H2	P 3.1 · L 5.5 · R 5.59	INTERP MED · P-gap 1g · C6
H3	P 8.11 · L 6.7 · R 6.7	INTERP MED · P-gap 1g · C20 · C5
SRU	P 9.53 · L 9.4 · R 9.53	INTERP MED · P-gap 4g · C20 · C-NWB-05 · C-T6
UIU	P 10.46 · L 10.6 · R 10.55	INTERP MED · P-gap 2g · C-NWB-01 · C3
LIU	P 11.63 · L 12.0 · R 11.9	INTERP LOW-MED · P-gap 1g · C4
DRU	P 14.2 · L 15.4 · R 14.91	INTERP MED · P-gap 2g · C20 · C1 · C9
MMU	P 16.0 · L 16.0 · R 16.0	INTERP LOW-MED · P-gap 1g · C9
BMU	P 23.13 · L 21.0 · R 21.0	INTERP LOW-MED · P-gap 1g · C7
BU	P 30.8 · L 30.8 · R 30.8	INTERP LOW-MED · P-gap 0g
LEU	P 34.1 · L 34.4 · R 34.4	INTERP LOW-MED · P-gap 1g · C11
ROU	P 56.0 · L 56.0 · R 56.0	INTERP LOW-MED · P-gap 1g · C-NWB-04

Restoration surfaces (REST15): H50 2.6 · H75 3.6 · H110 8.1 · H150 9.0 · H300 10.1 · H350 11.1 · H370 13.0 Ma — all INTERP LOW-MED, pack quotations, not governed ages.

WELL TIES · KB-KL2 AND REGIONAL CALIBRATION

C20: failed ties stay local — NOT propagated regionally

KB-KL2 WELLS (on the local screenshot)

ROTAN-1
Compiled NN: NN11 / NN10 · checkshot/VSP
SEABED ◆ EDGE · SRU × FAIL · H3 × FAIL

MALIGAN-1
Compiled NN: NN11 / NN10 / NN9 · checkshot/VSP
SRU × FAIL · H3 ? UNKNOWN

BARTON-2
No NN control · checkshot/VSP

Shared object: WELL-ROTAN-1 (5 km off BIF-KL6 as tabulated)
KB-KL2 position on regional transect: navigation UNVERIFIED

REGIONAL CALIBRATION WELLS (compiled NN, SRC-PEN)

GAJAH HITIGA · TIGA PAPAN · MALIGAN · SOLISIP
GLAYZER-3 · RUSA TIMUR · IMBOK · BULUH
DELUAR · KETAM · SOUTH LOKA

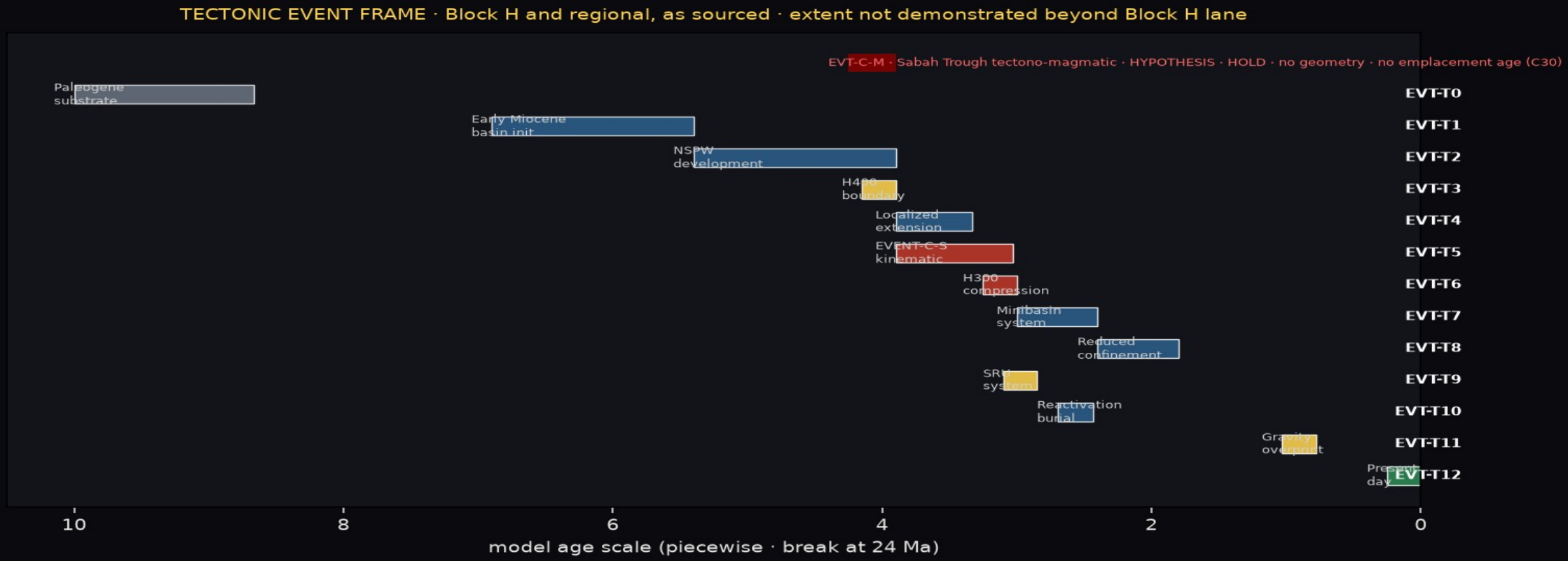
Not on KB-KL2 · C12

Gap grammar applied:
Erosion — not drawn (no source asserts located truncation)
Non-deposition — not drawn
Hiatus unresolved: UNKNOWN: EROSION OR NON-DEPOSITION
CONDENSATION CANDIDATE (inferred only)
Poor recovery / caving candidate (C12)
Reworking candidate (C12)

RC-04 · Well stick-to-name ambiguity on BIF-KL6 — three sticks under labels Buluh + Rotan (px 1011/1019/1024) and three under Kandi + Sugut (px 1398/1420/1429). OPEN.
RC-01 · KL2 marker on BIF-KL6 is BIF-KL2, not KB-KL2 — under C27 the printed 'KL2' at the 2W block (trough) is BIF-KL2; KB-KL2 relates to BIF-KL6 only through Rotan-1. OPEN · ARIF_REVIEW.

TECTONIC EVENT FRAME

All events: Block H and regional as sourced · extent NOT demonstrated beyond Block H lane · every event is INTERPRETATION · PARTIAL



EVENT-C SCOPE — EVENT-C-S: localized H370–H300 kinematic reorganisation in the Tanjung / Rotan restoration corridor (Block H, outboard). Magmatism not shown.
EVENT-C-M: Sabah Trough tectono-magmatic branch — HYPOTHESIS · HOLD · no geometry · no emplacement age (C30). Neither explains the regional transect.
Competing models for EVENT-C-S: syn-compressional gravitational collapse · back-limb extension · strain partitioning · oblique reactivation · mobile-shale decoupling · restoration artefact.

EVENT DETAIL · PALEO SYSTEM CONSEQUENCE

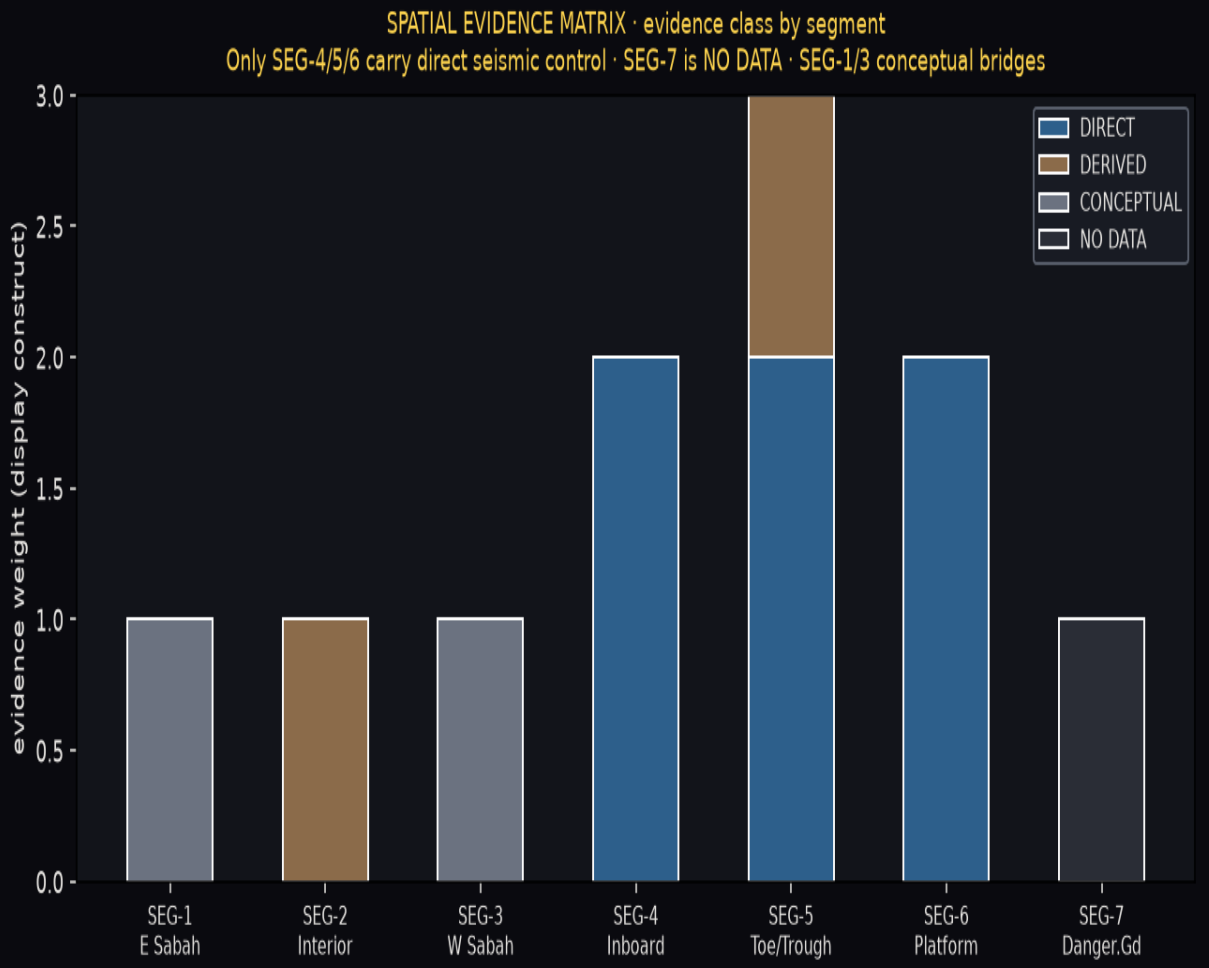
Every row: evidence class = INTERPRETATION · PARTIAL · open gates G4/G5 (G6 for Event C)

EVT-T0	Paleogene inherited substrate	Deep surfaces NP14 46.29 / NP8 57.21 Ma model-assigned	Substrate controls heat flow & deep kitchens · unresolved	G4,G5
EVT-T1	Early Miocene basin initiation	RELQ25 p.22 · how NSPW emplaced above older prism (C-T3)	Future mobile-shale supply · NSPW a theoretical source	G4,G5
EVT-T2	NSPW development	circa 18–13 Ma (RELQ25 p.22) · composite NSPW origin (C-T4)	Overpressure potential (analogue)	G4,G5
EVT-T3	H400 structural boundary	Age NOT assigned · pack labels H400 and H370 both 13 Ma (REST15)	Deep kitchens and conduits below H400 poorly imaged	G4,G5
EVT-T4	Localized extension during shortening	13 → 11.1 Ma · collapse / shale-withdrawal / thrust-top? (C-T7)	This depocentre later becomes the Tanjung kitchen	G4,G5
EVT-T5	EVENT-C-S kinematic reorganisation	13 → 10.1 Ma · not mapped to stored surfaces · link to Trough? (C26)	Alters routing of later sands and kitchen geometry	G4,G5
EVT-T6	H300 compressional pulse	10.1 Ma restored → ~10 Ma deformed · detachment geometry open	Rotan trap forms	G4,G5
EVT-T7	Confined-minibasin system	No age assigned · minibasin mechanism (C-T7) · mud genesis (C-T14)	Ponded reservoirs · mud-bounded traps	G4,G5
EVT-T8	Reduced regional confinement	No age assigned · where and why confinement persisted open	Fairways must stay horizon-specific	G4,G5
EVT-T9	SRU surface system	P 9.53 / L 9.4 / R 9.53 Ma · pack-marker equivalence CORRELATION HOLD	Shelf foreset pay · deepwater expression not erosional	G4,G5
EVT-T10	Local reactivation and burial	9.0 → 8.1 Ma · spatial extent of reactivation open	Trap modification · possible breach	G4,G5
EVT-T11	Gravity-dominated overprint	H50 2.6 Ma · Stage IVF (RELQ25 p.24) · gravity vs compression (C-T10)	Hikmat-1: 4.6 m net gas in H50	G4,G5
EVT-T12	Present day	Stress tensor on KL2 UNKNOWN · fluid escape not demonstrated	Partial leakage · most accumulations not filled to spill	G4,G5

H400 AGE · GOVERNED PAYLOAD: UNASSIGNED · REST15 QUOTATIONS: 13 Ma · STATUS: HOLD · ARIF DECISION: UNSET

SPATIAL EVIDENCE MATRIX

Segment widths NOT to a common scale · SEG boundaries are GEOX display constructs (RG-01)



KEY GATES OPEN

RG-01 Segment display boundaries on frame A–A′
RG-02 Designate DEM topographic source
RG-03 Freeze BIF-KL6 identity and navigation
RG-04 NW Sabah Platform vs Dangerous Grounds boundary
RG-05 Designate East Sabah geological-map control
RG-06 Carry or reject operator 'Mesozoic' attribution
RG-07 Crustal control for R1–R4 (wide-angle / MT)

C9 MMU vs DRU across Sabah Trough — kept separate
C20 Failed KB-KL2 ties stay local
C27 BIF-KL6 namespace: REGIONAL-COMPOSITE-KL6
C30 EVENT-C-M: no geometry, no emplacement age
RC-06 Kinabalu root / igneous corridor NOT adopted
RC-07 Public-frame Moho depths NOT adopted

All decisions: UNSET. GEOX recommends nothing as closed.

EPISTEMIC SEPARATION · WHAT IS KNOWN WHERE

Build correctness is not Earth truth. GEOX renders; ARIF adjudicates.

EPISTEMIC SEPARATION · what is known where · GEOX renders; ARIF adjudicates

WHAT THE EVIDENCE DIRECTLY SUPPORTS
BIF-KL6 composite: rugose platform seabed · flat trough floor ≈3800 ms · seabed rising SE · offset reflectors in toe-thrust panel

WHAT IS LOCAL TO KB-KL2
Digitised horizons · Rotan-1/Maligan-1/Barton-2 ties (C20 FAIL) · H-series (H400 age unassigned) · restoration chronology behind EVENT-C-S

WHAT MAY BE REGIONAL
Shallowing-to-SE seabed trend · trough → toe-thrust → thrust-sheet → shelf domain order · consistent with public A–A′ frame order

WHAT REMAINS A CORRELATION
DRU/SRU/H-series continuity KB-KL2 → BIF-KL6 · MMU vs DRU across trough (C9) · KL6 panel positions relative to frame A–A′

WHAT REMAINS A HYPOTHESIS
EVENT-C-M · crustal models R1–R4 · Models A/B/C (SRC-ESDG-MODELS) · onshore–offshore magmatic correlation · 'Mesozoic' attribution of deep SE section

NOT REPRESENTED BY DATA: East Sabah & interior subsurface · mapped surface geology along transect · Dangerous Grounds below seabed · crust and Moho everywhere · depth everywhere

ARIF REGIONAL REVIEW QUEUE

geox_sabah_regional_v1 · all decisions UNSET · GEOX recommends nothing as closed

1	**RG-01**	Approve GEOX segment display boundaries (SEG-1…SEG-7 km ranges)	ARIF confirmation or replacement transect polyline
2	**RG-02**	Designate a topographic source (DEM e.g. GEBCO/SRTM) to replace schematic nodes	approved DEM extraction along the transect
3	**RG-03**	Freeze BIF-KL6 identity and navigation; fix well positions	SEG-Y / navigation, well header coordinates (SRC-SABAH-WSM)
4	**RG-04**	NW Sabah Platform vs Dangerous Grounds boundary and naming	regional province map
5	**RG-05**	Designate East Sabah geological-map control (SB505 MR2 / Sandakan / other)	ARIF designation; then ingestion audit
6	**RG-06**	Carry or reject the operator 'Mesozoic' attribution on BIF-KL6	well / outcrop age control
7	**RG-07**	Crustal control for R1–R4 (wide-angle, refraction, MT)	geophysical acquisition or published constrained model
8	**ARIF-H400-AGE**	H400 age: governed UNASSIGNED vs REST15 13 Ma · HOLD	adjudicate 13 Ma REST15 quotation vs unassigned governed age
9	**C27 / RC-01**	Confirm BIF-KL6 namespace; printed 'KL2' marker is BIF-KL2	line identity
10	**C9**	MMU vs DRU across the Sabah Trough (kept separate)	dip keyline across DG platforms
11	**C20**	Failed KB-KL2 ties stay local (no regional propagation)	VSP, navigation, DSG export
12	**RC-06**	Keep Kinabalu root / igneous corridor out of the governed model	cited constrained model

NEW CONTRADICTIONS: RC-01 KL2 marker identity · RC-02 KB-KL2 position unknown · RC-03 Platform vs Dangerous Grounds province · RC-04 well stick-to-name ambiguity · RC-05 'Mesozoic' outside 0–66 Ma chart · RC-06 Kinabalu root NOT adopted · RC-07 public Moho NOT adopted.

HIGHEST INFORMATION VALUE DATA · AND WHAT IT CHANGES

WHAT DATA HAVE THE HIGHEST INFORMATION VALUE

1) Navigated SEG-Y of BIF-KL6 (or equivalent dip line) with well ties → resolves C9 and the trough-to-shelf correlation

2) Survey IDs and header coordinates for the KL6 wells (SRC-SABAH-WSM lists several) → resolves RC-04, RG-03

3) Wide-angle / MT along the margin for R1–R4 → resolves RG-07

4) Digitised geological map for SEG-1 / SEG-2 → resolves RG-05

WHAT CHANGES EXPLORATION UNDERSTANDING

KB-KL2 cannot be used as the regional reference: its failed ties stay local.

Regional play statements need a navigated dip keyline across the trough tied to wells — until then regional maps should carry CORRELATION, not TIE, status.

(Consequence statements are interpretation, not fact.)

SOURCES

SRC-XS-PUB (public schematic frame + place coordinates) · SRC-BIF23-KL6 (NWB23 deck seismic display raster) · kinabalu_chronostrat_v2.json (scale, GTS2020, NN, megasequences, regional wells) · v4.11 RHG payload (surfaces, ties, gates, Wheeler) · v3.1 surface system (operator ages, SRC-REST15) · SRC-ESDG-MODELS (hypotheses) · SRC-RELQ25 · SRC-REP15 · SRC-NWB23 · SRC-ARIF-EVC · SRC-ENT-LL1

Inventoried, NOT ingested: SB505 MR2 · Sandakan publication · SB304 deck. No external calls. No image generation. Base chart: rhg_v4_2_chronostrat KL2_v4_2_CHRONOSTRAT.svg embedded unchanged (hash-checked).